

IOT BASED HEART MONITORING AND RISK PREDICTION

by

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2025-2026

Title Explanation

IoT Based Heart Monitoring and Risk Prediction

IoT-Based System

Integrates biomedical sensors with an ESP32 to enable real-time data acquisition and wireless transmission.

Heart Monitoring

Tracks key cardiac parameters including heart rate, SpO₂, and ECG-derived RR intervals.

Risk Prediction

Uses predefined medical thresholds to evaluate and classify cardiac health status.

Data Visualization

Presents health trends through cloud-based dashboards for daily, weekly, and monthly monitoring.

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Abstract

Introduction

Continuous monitoring of cardiac and vascular diseases is essential for early detection of heart abnormalities and preventive healthcare.

Research Gap

Existing healthcare and wearable systems are cost-intensive, non portable, and lack integrated automatic risk classification and long-term data visualization.

Methodology

An IoT-based system using MAX30100 and AD8232 sensors with an ESP32 microcontroller collects physiological data and transmits data to the cloud.

Results

Physiological parameters are classified into low, medium, and high cardiac risk levels and cloud-based storage for continuous monitoring.

Conclusion

The proposed system provides a low-cost, scalable solution with effective risk classification and dashboard-based visualization for early identification of cardiac abnormalities.

Introduction

Problem Identified

Existing cardiac monitoring systems are expensive, hospital-dependent, and unsuitable for continuous remote monitoring.

Why This Project

To design a **low-cost, portable, and IoT-enabled solution** for continuous heart health monitoring outside clinical environments.

What This Project Does

Measures **heart rate, SpO₂, and ECG-based RR interval**, classifies cardiac risk, and visualizes trends using a cloud dashboard.

Why This Approach

Focuses on **simplicity, real-time monitoring, and interpretability** using lightweight and interpretable machine learning model.



Software Requirements

Development Tools

- Arduino IDE
- ESP32 Board Package (Espressif Systems)
- Google Apps Script
- Google Sheets (Cloud Platform)

Programming Languages

- Embedded C++ (Arduino framework)
- JavaScript (Google Apps Script)
- Python (Machine Learning & Data Processing)

IoT Products Used

- ESP32 Microcontroller
- MAX30100 Pulse Oximeter Sensor
- AD8232 ECG Sensor Module

IoT Platforms & Libraries

- Wi-Fi & HTTP client Libraries (ESP32)
- MAX30100 PulseOximeter Library
- Wire (I²C) Library

Datasets/Modules

Dataset Description

- Physiological data is collected from MAX30100 and AD8232 sensors during real-time monitoring.
- Historical sensor readings are stored in Google Sheets and organized as a structured dataset.
- Predefined threshold ranges are used to assign risk labels to the collected parameters.
- The labeled dataset is used to train the decision tree–based risk classification model.
- New incoming sensor data is classified using the trained model and stored for visualization and analysis.



Methodology

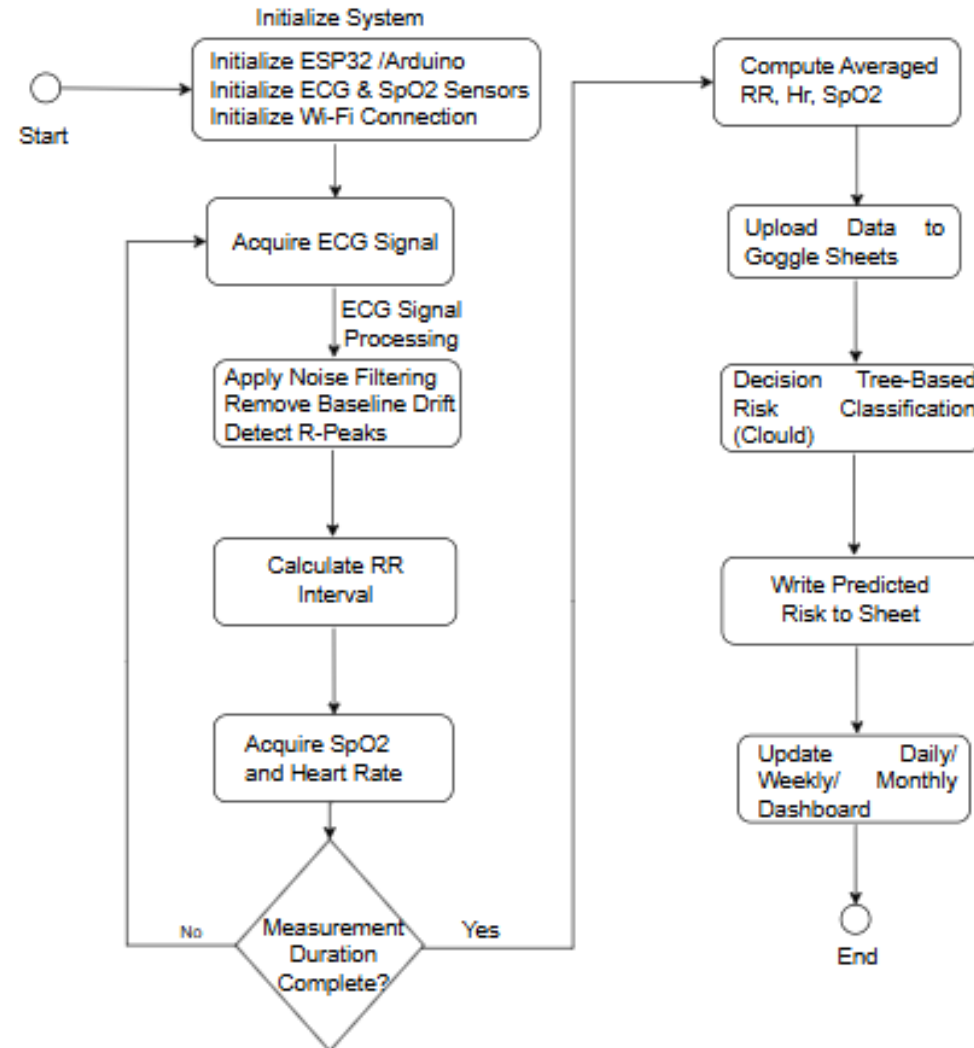
- Physiological signals (Heart Rate, SpO₂, ECG) are acquired using MAX30100 and AD8232 sensors.
- The ESP32 microcontroller performs signal preprocessing and extracts relevant parameters such as HR, SpO₂, and RR interval.
- Processed data is transmitted wirelessly to the cloud using the ESP32 Wi-Fi module.
- Cloud storage (Google Sheets) is used to store and manage time-stamped health data.
- A decision tree-based machine learning model is used to classify cardiac risk into Low, Medium, or High levels.
- Dashboard visualization presents daily, weekly, and monthly health trends for continuous monitoring.



Process Flow



UML Activity Diagram



Implementation

Code Snippet 1: Sensor Initialization & Libraries

```
#include <WiFi.h>
#include <HTTPClient.h>
#include <Wire.h>
#include "MAX30100_PulseOximeter.h"

PulseOximeter pox;

/* ----- PINS ----- */
#define ECG_PIN 34
#define LO_PLUS 32
#define LO_MINUS 33
```

Code Snippet 2: ECG Signal Processing_toggle_baseline removal

```
sum -= buffer[idx];
buffer[idx] = raw;
sum += raw;
idx = (idx + 1) % MA_WINDOW;
float smooth = sum / (float)MA_WINDOW;

baseline = alpha * baseline + (1 - alpha) * smooth;
float ecg = smooth - baseline;
```

Implementation

Code Snippet 3: Data Upload to Google Sheets

```
String url = GOOGLE_SCRIPT_URL +  
            "?rr=" + String(rr) +  
            "&hr=" + String(hr, 1) +  
            "&spo2=" + String(spo2, 1);  
http.begin(url);  
http.GET();  
http.end();
```

Code Snippet 4: Cloud-Based Decision Tree Classification

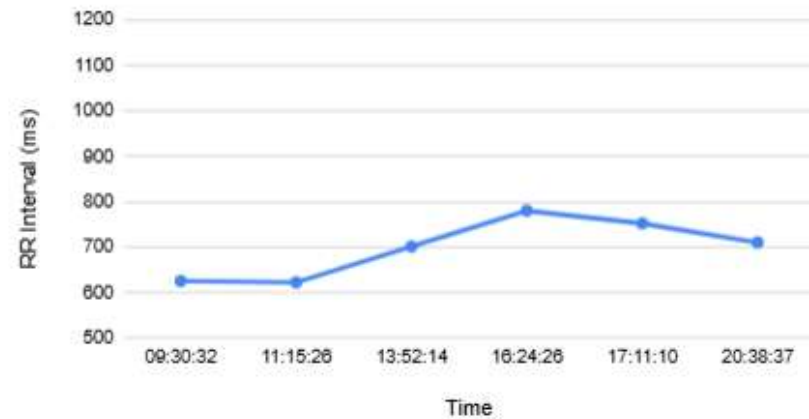
```
# Load training data from Google Sheets  
train_df = pd.DataFrame(train_sheet.get_all_records())  
  
X = train_df[["Avg_RR(ms)", "Avg_HR(bpm)", "Avg_SpO2(%)"]]  
y = train_df["Final"]  
  
# Encode labels  
label_encoder = LabelEncoder()  
y_encoded = label_encoder.fit_transform(y)  
  
# Train Decision Tree model  
model = DecisionTreeClassifier(max_depth=4)  
model.fit(X, y_encoded)  
  
# Predict risk level for new sensor data  
prediction = model.predict(new_data)  
risk_level = label_encoder.inverse_transform(prediction)
```

Results

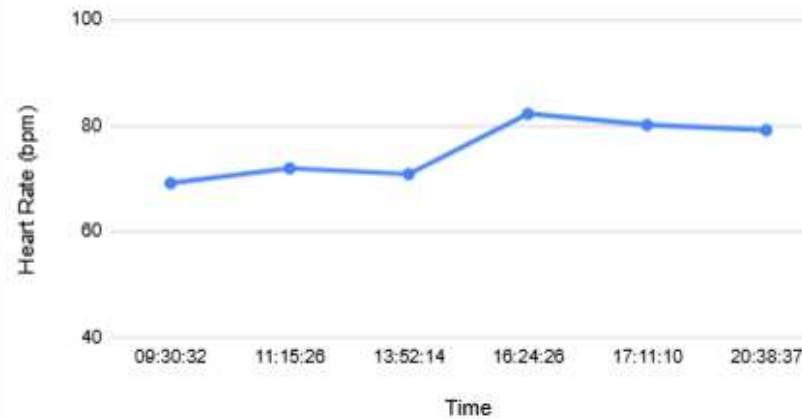
Current Health Status

LOW

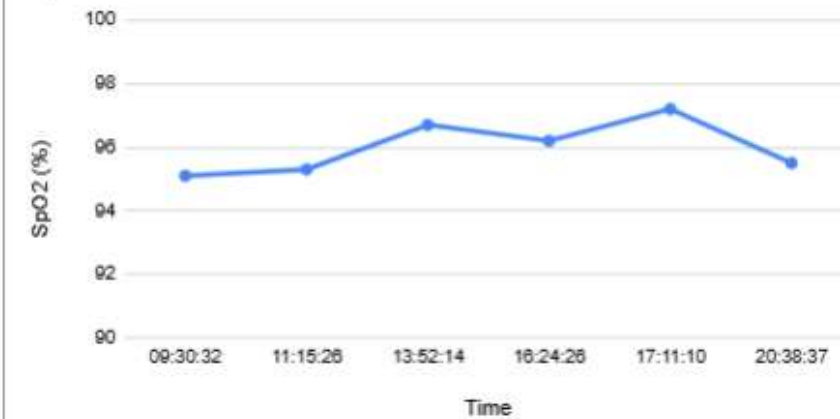
RR Trends



HR Trends



SpO2 Trends



Status interpretation based on RR, HR, and SpO2 thresholds

LOW:

LOW indicates vital parameters are within normal physiological range

MEDIUM:

MEDIUM indicates moderate deviation in vital parameters requiring observation

HIGH:

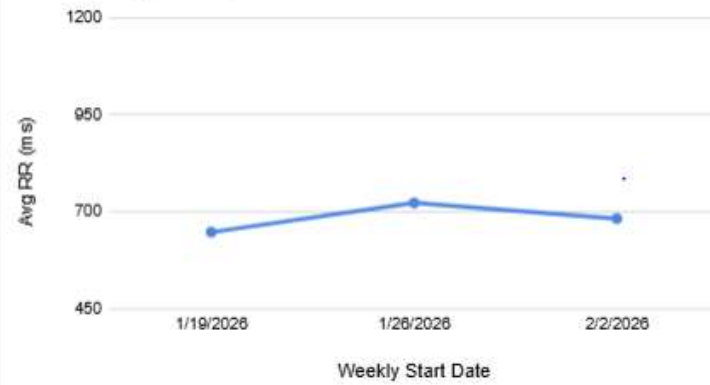
HIGH indicates critical deviation in vital parameters requiring immediate attention

Results

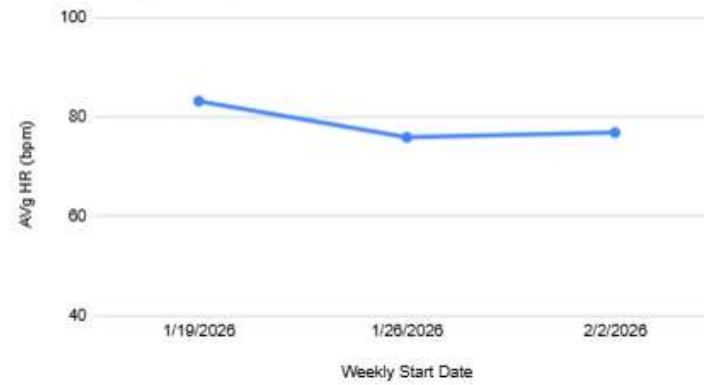
Current Week Health Status

LOW

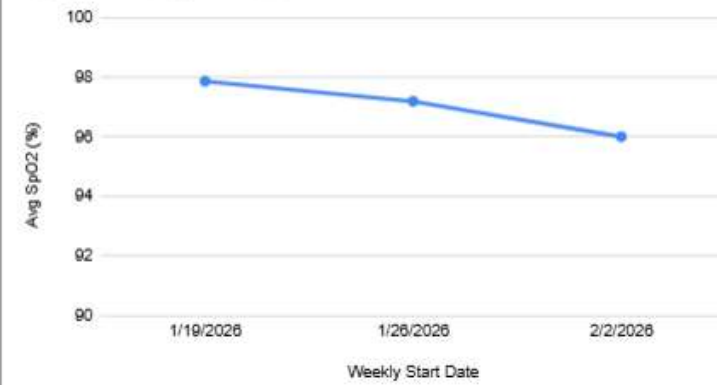
RR Weekly Trends



HR Weekly Trends



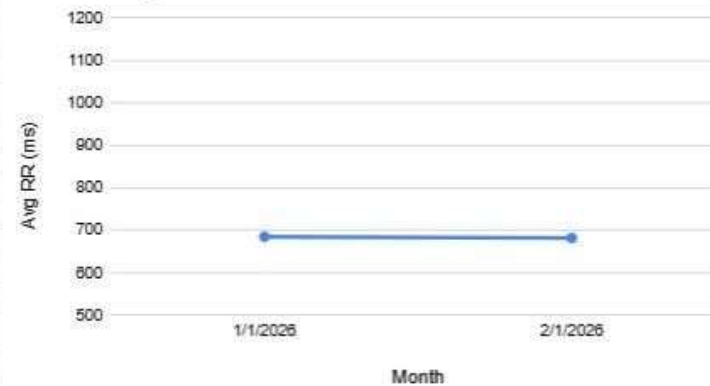
SpO2 Weekly Trends



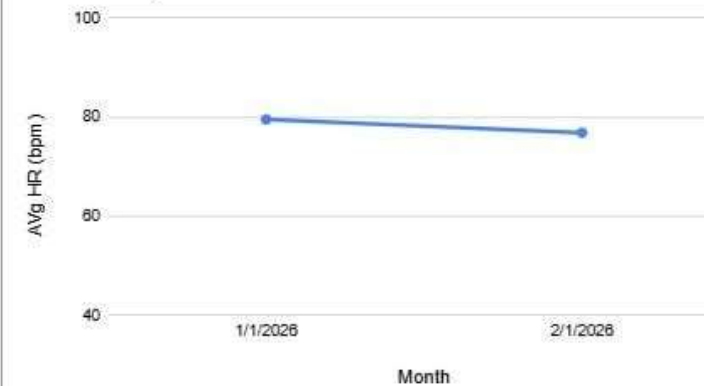
Current Month Health Status

LOW

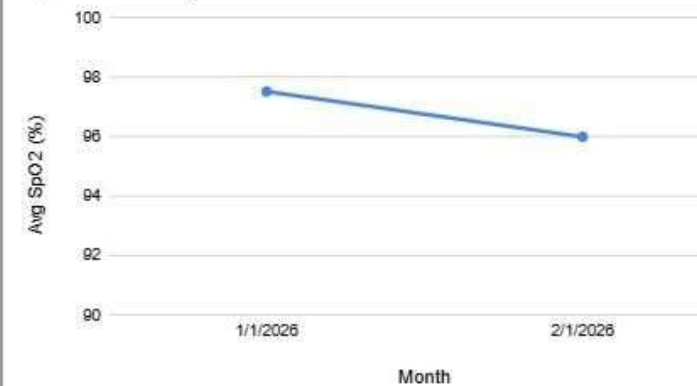
RR Monthly Trends



HR Monthly Trends



SpO2 Monthly Trends



Conclusion

Key Achievements

- 1
 - Developed a low-cost IoT-based system for continuous monitoring of HR, SpO₂, and RR interval.
 - Implemented real-time signal processing and decision tree-based cardiac risk classification.
 - Cloud integration enables remote access, data storage, and trend visualization.

Future Enhancements

- 2
 - Future work includes real-time alerts (SMS/app) for high-risk conditions.

System Extensibility

- 3
 - System can be extended with advanced ML models, more sensors, and a mobile application.

References

Journal Articles

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- [2] B. Lucas, A. Mahamuni, V. Kulal, A. Gupta, and K. Deorukhkar, “Affordable real-time heart rate, ECG, and SpO₂ monitoring system using Internet of Things,” *International Journal for Research in Applied Science & Engineering Technology*, vol. 10, no. 11, pp. 1–8, Nov. 2022, doi: 10.22214/ijraset.2022.47731.
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- [4] N. Divya, M. Riyazuddin, A. Ahad, S. R. Vulapula, A. Manjula, and M. Sirajuddin, “Predicting heart disease using machine learning and IoT techniques,” *Onkologia i Radioterapia*, vol. 18, no. 9, pp. 001–011, Nov. 2024.
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Thank You



Any Queries

